1) a)

$2 \times (3) + 8 \times (1) - 3 \times \left(\frac{10}{3}\right) = 6 + 8 - 10 = 4$

So $\left(3, 1, \frac{10}{3}\right)$ lies on the plane Π

1) b)

$2 \times (21) + 8 \times (-2) - 3 \times (7) = 42 - 16 - 21 = 5 \neq 4$

So $(21, -2, 7)$ does not lie on the plane Π

2) a)

$2(3) + (-5) + 6(4) = 6 - 5 + 24 = 25 \neq 24$

So A does not lie on the plane Π_1

2) b)

$-5(3) - 2(-5) + 3(4) = -15 + 10 + 12 = 7$

So A does lie on the plane Π_2

2) c)

$8(3) + 3(-5) - 8(4) = 24 - 15 - 32 = -23$

So A does lie on the plane Π_3

2) d)

$\frac{1}{2}(3) + \frac{3}{2}(-5) + 3(4) = \frac{3}{2} - \frac{15}{2} + 12 = 6$

So A does lie on the plane Π_4

3) a)

$$\begin{pmatrix} 3 \\ -5 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -16 \\ -7 \\ 13 \end{pmatrix} = 3 \times (-16) + (-5) \times (-7) + 1 \times 13 = 0$$

So the plane Π contains the vector $\begin{pmatrix} -16 \\ -7 \\ 13 \end{pmatrix}$

3) b)

$$\begin{pmatrix}3\\-5\\1\end{pmatrix}\cdot\begin{pmatrix}13\\4\\-19\end{pmatrix}=3\times13+(-5)\times4+1\times(-19)=0$$

So the plane Π contains the vector $\begin{pmatrix}13\\4\\-19\end{pmatrix}$

3) c)

$$\begin{pmatrix}3\\-5\\1\end{pmatrix}\cdot\begin{pmatrix}-13\\-7\\-16\end{pmatrix}=3\times(-13)+(-5)\times(-7)+1\times(-16)=-20\neq0$$

So the plane Π does not contain the vector $\begin{pmatrix}-13\\-7\\-16\end{pmatrix}$

4) a)

The plane has equation $2x+2y+z=d$

$d=2(1)+2(4)+1=11$

$\therefore 2x+2y+z=11$

4) b)

The plane has equation $-x+4y-3z=d$

$d=-(6)+4(1)-3(-2)=4$

$\therefore -x+4y-3z=4$

4) c)

The plane has equation $5x-y-2z=d$

$d=5(-3)-(-4)-2(-9)=7$

$\therefore 5x-y-2z=7$

4) d)

The plane has equation $-2x+3y+10z=d$

$d=-2\left(\frac{1}{2}\right)+3\left(-\frac{3}{2}\right)+10\left(\frac{5}{2}\right)=\frac{39}{2}$

$\therefore -2x+3y+10z=\frac{39}{2}$ or $-4x+6y+20z=39$

4) e)

The plane has equation $0x + 7y - 8z = d$

$d = 0(3) + 7(0) - 8(-2) = 16$

$\therefore 7y - 8z = 16$

4) f)

The plane has equation $-4x + 15y + 0z = d$

$d = -4\left(-\frac{1}{3}\right) + 15\left(\frac{1}{4}\right)0\left(\frac{2}{3}\right) = \frac{61}{12}$

$\therefore -4x + 15y = \frac{61}{12}$ or $-48x + 180y = 61$

5) a)

The plane has equation $5x - 2y + 4z = d$

$d = 5(2) - 2(1) + 4(3) = 20$

$\therefore 5x - 2y + 4z = 20$

5) b)

$B(3, 1, -5)$:

$5(3) - 2(1) + 4(-5) = -7 \neq 20$

So B does not lie on the plane.

$C(-2, 0, 8)$:

$5(-2) - 2(0) + 4(8) = 22 \neq 20$

So C does not lie on the plane.

6)

The plane has equation $4x - y + z = d$

$d = 4(1) - (1) + (4) = 7$

$\therefore 4x - y + z = 7$

7)

The plane has equation $-3x + 8y - z = d$

$d = -3(2) + 8\left(\frac{1}{2}\right) - (-8) = 6$

$\therefore -3x + 8y - z = 6$

8)

The plane has equation $kx + 3y - 2kz = d$

$d = k(3) + 3(-4k) - 2k(5) = 3k - 12k - 10k = -19k$

$\therefore kx + 3y - 2kz = -19k$

9) a)

$\begin{pmatrix} -1 \\ 6 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ p \end{pmatrix} = 0 \Rightarrow -1 \times 1 + 6 \times 1 + (-2) \times p = 0 \Rightarrow 5 - 2p = 0 \Rightarrow p = \frac{5}{2}$

9) b)

The plane has equation $x + y + \frac{5}{2}z = d$

$d = 1(2) + 1(1) + \frac{5}{2}(1) = \frac{11}{2}$

$\therefore x + y + \frac{5}{2}z = \frac{11}{2}$

$\Rightarrow 2x + 2y + 5z = 11$

10)

$\begin{pmatrix} k \\ -6 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 8 \\ -3 \end{pmatrix} = 0 \Rightarrow k \times 4 + (-6) \times 8 + 1 \times (-3) = 0$

$\Rightarrow 4k - 48 - 3 = 0 \Rightarrow 4k - 51 = 0 \Rightarrow k = \frac{51}{4}$

11)

$\begin{pmatrix} 5 \\ 1 \\ -k \end{pmatrix} \cdot \begin{pmatrix} 2k \\ -7 \\ 2 \end{pmatrix} = 0 \Rightarrow 5 \times (2k) + 1 \times (-7) + (-k) \times 2 = 0$

$\Rightarrow 10k - 7 - 2k = 0 \Rightarrow 8k - 7 = 0 \Rightarrow k = \frac{7}{8}$

12)

$$\begin{pmatrix} k+1 \\ -2 \\ 3k \end{pmatrix} \cdot \begin{pmatrix} 4k \\ -(k+2) \\ -10 \end{pmatrix} = 0 \Rightarrow (k+1) \times 4k + (-2) \times \big(-(k+2)\big) + 3k \times (-10) = 0$$

$$\Rightarrow 4k^2 + 4k + 2k + 4 - 30k = 0$$

$$\Rightarrow 4k^2 - 24k + 4 = 0$$

$$\Rightarrow k^2 - 6k + 1 = 0$$

$$\Rightarrow (k-3)^2 - 9 + 1 = 0$$

$$\Rightarrow (k-3)^2 = 8$$

$$\Rightarrow k = 3 \pm 2\sqrt{2}$$

13) a)

First find the angle between the vector and the normal to the plane:

$$\cos\theta = \frac{\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}}{\sqrt{1^2+2^2+4^2}\sqrt{1^2+4^2+3^2}}$$

$$\Rightarrow \cos\theta = \frac{1\times1+2\times4+4\times3}{\sqrt{21}\times\sqrt{26}}$$

$$\Rightarrow \cos\theta = \frac{\sqrt{546}}{26} \Rightarrow \theta = 26.010\ldots°$$

So angle between the vector and the plane is:

Angle $= 90° - 26.010\ldots° = 64.0°$ (1 DP)

13) b)

First find the angle between the vector and the normal to the plane:

$$\cos\theta = \frac{\begin{pmatrix} -1 \\ 3 \\ -8 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}}{\sqrt{(-1)^2+3^2+(-8)^2}\sqrt{2^2+(-1)^2+2^2}}$$

$$\Rightarrow \cos\theta = \frac{(-1)\times2+3\times(-1)+(-8)\times2}{\sqrt{74}\times\sqrt{9}}$$

$$\Rightarrow \cos\theta = -\frac{7\sqrt{74}}{74} \Rightarrow \theta = 144.462\ldots°$$

Acute Angle $= 180° - 144.462\ldots° = 35.537\ldots°$

So angle between the vector and the plane is:

Angle $= 90° - 35.537\ldots° = 54.5°$ (1 DP)

13) c)

First find the angle between the vector and the normal to the plane:

$$\cos\theta = \frac{\begin{pmatrix}1\\8\\0\end{pmatrix}\cdot\begin{pmatrix}5\\0\\-4\end{pmatrix}}{\sqrt{1^2+8^2+0^2}\sqrt{5^2+0^2+(-4)^2}}$$

$$\Rightarrow \cos\theta = \frac{1\times5+8\times0+0\times(-4)}{\sqrt{65}\times\sqrt{41}}$$

$$\Rightarrow \cos\theta = \frac{\sqrt{2665}}{533} \Rightarrow \theta = 84.441\ldots°$$

So angle between the vector and the plane is:

Angle $= 90° - 84.441\ldots° = 5.6°$ (1 DP)

13) d)

First find the angle between the vector and the normal to the plane:

$$\cos\theta = \frac{\begin{pmatrix}9\\0\\-3\end{pmatrix}\cdot\begin{pmatrix}-1\\3\\2\end{pmatrix}}{\sqrt{9^2+0^2+(-3)^2}\sqrt{(-1)^2+3^2+2^2}}$$

$$\Rightarrow \cos\theta = \frac{9\times(-1)+0\times3+(-3)\times2}{\sqrt{90}\times\sqrt{14}}$$

$$\Rightarrow \cos\theta = -\frac{\sqrt{35}}{14} \Rightarrow \theta = 114.997\ldots°$$

Acute Angle $= 180° - 114.997\ldots° = 65.002\ldots°$

So angle between the vector and the plane is:

Angle $= 90° - 65.002\ldots° = 25.0°$ (1 DP)

14) a)

The XY plane has equation $z = 0$

$$\cos\theta = \frac{\begin{pmatrix}1\\-1\\-2\end{pmatrix}\cdot\begin{pmatrix}0\\0\\1\end{pmatrix}}{\sqrt{1^2+(-1)^2+(-2)^2}\times\sqrt{0^2+0^2+1^2}}$$

$$\Rightarrow \cos\theta = \frac{1\times0+(-1)\times0+(-2)\times1}{\sqrt{6}\times\sqrt{1}}$$

$$\Rightarrow \cos\theta = -\frac{2}{\sqrt{6}} \Rightarrow \theta = 144.735\ldots°$$

Acute Angle $= 180° - 144.735\ldots° = 35.3°$ (1 DP)

14) b)

The XZ plane has equation $y = 0$

$$\cos\theta = \frac{\begin{pmatrix}1\\-1\\-2\end{pmatrix}\cdot\begin{pmatrix}0\\1\\0\end{pmatrix}}{\sqrt{1^2+(-1)^2+(-2)^2}\times\sqrt{0^2+1^2+0^2}}$$

$$\Rightarrow \cos\theta = \frac{1\times0+(-1)\times1+(-2)\times0}{\sqrt{6}\times\sqrt{1}}$$

$$\Rightarrow \cos\theta = -\frac{1}{\sqrt{6}} \Rightarrow \theta = 114.094\ldots°$$

Acute Angle $= 180° - 114.094\ldots° = 65.9°$ (1 DP)

14) c)

The YZ plane has equation $x = 0$

$$\cos\theta = \frac{\begin{pmatrix}1\\-1\\-2\end{pmatrix}\cdot\begin{pmatrix}1\\0\\0\end{pmatrix}}{\sqrt{1^2+(-1)^2+(-2)^2}\times\sqrt{0^2+1^2+0^2}}$$

$$\Rightarrow \cos\theta = \frac{1\times1+(-1)\times0+(-2)\times0}{\sqrt{6}\times\sqrt{1}}$$

$$\Rightarrow \cos\theta = \frac{1}{\sqrt{6}} \Rightarrow \theta = 65.905\ldots°$$

Acute Angle $= 65.9°$ (1 DP)

15)

$$\cos\theta = \frac{\begin{pmatrix}2\\3\\-8\end{pmatrix}\cdot\begin{pmatrix}0\\0\\1\end{pmatrix}}{\sqrt{2^2+3^2+(-8)^2}\times\sqrt{0^2+0^2+1^2}}$$

$$\Rightarrow \cos\theta = \frac{2\times0+3\times0+(-8)\times1}{\sqrt{77}\times\sqrt{1}}$$

$$\Rightarrow \cos\theta = -\frac{8}{\sqrt{77}} \Rightarrow \theta = 155.739\ldots°$$

Acute Angle $= 24.260\ldots° = 24.3°$ (1 DP)

A 20% incline is an angle $\phi = \arctan\frac{20}{100} = 11.3°$ (1 DP)

$24.3° > 11.3°$ so the new road is too steep and will be rejected.

16) a)

$$\cos\theta = \frac{\begin{pmatrix}1\\1\\-1\end{pmatrix}\cdot\begin{pmatrix}1\\1\\-2\end{pmatrix}}{\sqrt{1^2+1^2+(-1)^2}\times\sqrt{1^2+1^2+(-2)^2}}$$

$$\Rightarrow \cos\theta = \frac{1\times1+1\times1+(-1)\times(-2)}{\sqrt{3}\times\sqrt{6}}$$

$$\Rightarrow \cos\theta = \frac{4}{3\sqrt{2}} = \frac{2\sqrt{2}}{3} \Rightarrow \theta = 19.471\ldots°$$

Acute Angle $= 19.5°$ (1 DP)

16) b)

$$\cos\theta = \frac{\begin{pmatrix}3\\2\\1\end{pmatrix}\cdot\begin{pmatrix}8\\1\\2\end{pmatrix}}{\sqrt{3^2+2^2+1^2}\times\sqrt{8^2+1^2+2^2}}$$

$$\Rightarrow \cos\theta = \frac{3\times8+2\times1+1\times2}{\sqrt{14}\times\sqrt{69}}$$

$$\Rightarrow \cos\theta = \frac{28}{\sqrt{966}} = \frac{2\sqrt{966}}{69} \Rightarrow \theta = 25.725\ldots°$$

Acute Angle $= 25.7°$ (1 DP)

16) c)

$$\cos\theta = \frac{\begin{pmatrix}4\\-1\\-1\end{pmatrix}\cdot\begin{pmatrix}-1\\1\\1\end{pmatrix}}{\sqrt{4^2+(-1)^2+(-1)^2}\times\sqrt{(-1)^2+1^2+1^2}}$$

$$\Rightarrow \cos\theta = \frac{4\times(-1)+(-1)\times1+(-1)\times1}{\sqrt{18}\times\sqrt{3}}$$

$$\Rightarrow \cos\theta = -\frac{6}{3\sqrt{6}} = -\frac{\sqrt{6}}{3} \Rightarrow \theta = 144.735\ldots°$$

Acute Angle $= 180° - 144.735\ldots° = 35.3°$ (1 DP)

16) d)

$$\cos\theta = \frac{\begin{pmatrix}-5\\-2\\1\end{pmatrix}\cdot\begin{pmatrix}9\\10\\-6\end{pmatrix}}{\sqrt{(-5)^2+(-2)^2+1^2}\times\sqrt{9^2+10^2+(-6)^2}}$$

$$\Rightarrow \cos\theta = \frac{(-5)\times9+(-2)\times10+1\times(-6)}{\sqrt{30}\times\sqrt{217}}$$

$$\Rightarrow \cos\theta = -\frac{71}{\sqrt{6510}} = -\frac{71\sqrt{6510}}{6510} \Rightarrow \theta = 151.638\ldots°$$

Acute Angle $= 180° - 151.638\ldots° = 28.4°$ (1 DP)

16) e)

$$\cos\theta = \frac{\begin{pmatrix}1\\0\\-4\end{pmatrix}\cdot\begin{pmatrix}0\\1\\11\end{pmatrix}}{\sqrt{1^2+0^2+(-4)^2}\times\sqrt{0^2+1^2+11^2}}$$

$$\Rightarrow \cos\theta = \frac{1\times0+0\times1+(-4)\times11}{\sqrt{17}\times\sqrt{122}}$$

$$\Rightarrow \cos\theta = -\frac{44}{\sqrt{2074}} = -\frac{22\sqrt{2074}}{1037} \Rightarrow \theta = 165.051\ldots°$$

Acute Angle $= 180° - 165.051\ldots° = 14.9°$ (1 DP)

16) f)

$$\cos\theta = \frac{\begin{pmatrix}\sqrt{2}\\\sqrt{6}\\0\end{pmatrix}\cdot\begin{pmatrix}2\sqrt{2}\\2\sqrt{3}\\-1\end{pmatrix}}{\sqrt{\left(\sqrt{2}\right)^2+\left(\sqrt{6}\right)^2+0^2}\times\sqrt{\left(2\sqrt{2}\right)^2+\left(2\sqrt{3}\right)^2+(-1)^2}}$$

$$\Rightarrow \cos\theta = \frac{\sqrt{2}\times2\sqrt{2}+\sqrt{6}\times2\sqrt{3}+0\times(-1)}{\sqrt{8}\times\sqrt{21}}$$

$$\Rightarrow \cos\theta = \frac{4+6\sqrt{2}}{2\sqrt{42}} = \frac{\sqrt{42}+3\sqrt{21}}{21} \Rightarrow \theta = 15.579\ldots°$$

Acute Angle $= 15.6°$ (1 DP)

17)

$$\frac{\begin{pmatrix}1\\12\\0\end{pmatrix}\cdot\begin{pmatrix}1\\6\\k\end{pmatrix}}{\sqrt{1^2+12^2+0^2}\times\sqrt{1^2+6^2+k^2}} > \cos 5° = 0.996\ldots$$

$$\Rightarrow \frac{1\times1+12\times6+0\times k}{\sqrt{145}\times\sqrt{k^2+37}} > \cos 5°$$

$$\Rightarrow \frac{73}{\sqrt{145k^2+5365}} > \cos 5°$$

$$\Rightarrow \frac{73}{\cos 5°} > \sqrt{145k^2+5365}$$

$$\Rightarrow \frac{5329}{\cos^2 5°} > 145k^2+5365$$

$$\Rightarrow k^2 < \frac{5329}{145\cos^2 5°} - 37 = 0.0330316\ldots$$

$$\Rightarrow -0.182 < k < 0.182$$

18)

$$\frac{\begin{pmatrix}k\\1\\2\end{pmatrix}\cdot\begin{pmatrix}0\\20\\39\end{pmatrix}}{\sqrt{k^2+1^2+2^2}\times\sqrt{0^2+20^2+39^2}} < \cos 1° = 0.999847\ldots$$

$$\Rightarrow \frac{k\times 0+1\times 20+2\times 39}{\sqrt{k^2+5}\times\sqrt{1921}} < \cos 1°$$

$$\Rightarrow \frac{98}{\sqrt{1921k^2+9605}} < \cos 1°$$

$$\Rightarrow \frac{98}{\cos 1°} < \sqrt{1921k^2+9605}$$

$$\Rightarrow \frac{9604}{\cos^2 1°} < 1921k^2+9605$$

$$\Rightarrow k^2 > \frac{9604}{1921\cos^2 1°} - 5 = 0.001002675647\ldots$$

$\Rightarrow k < -0.0317$ (3 SF) or $k > 0.0317$ (3 SF)

19)

$$\cos 45° = \frac{\begin{pmatrix}6\\-1\\k\end{pmatrix}\cdot\begin{pmatrix}3\\-3\\0\end{pmatrix}}{\sqrt{6^2+(-1)^2+k^2}\times\sqrt{3^2+(-3)^2+0^2}}$$

$$\Rightarrow \frac{\sqrt{2}}{2} = \frac{6\times 3+(-1)\times(-3)+k\times 0}{\sqrt{37+k^2}\times\sqrt{18}}$$

$$\Rightarrow \frac{\sqrt{2}}{2} = \frac{21}{\sqrt{666+18k^2}}$$

$$\Rightarrow \sqrt{1332+36k^2} = 42$$

$$\Rightarrow 1332+36k^2 = 1764$$

$$\Rightarrow k^2 = 12 \Rightarrow k = \pm 2\sqrt{3}$$

As k is a positive constant, $k = 2\sqrt{3}$

20)

$$\cos 30° = \frac{\begin{pmatrix}9\\k\\-3\end{pmatrix}\cdot\begin{pmatrix}2\\0\\-2\end{pmatrix}}{\sqrt{9^2+k^2+(-3)^2}\times\sqrt{2^2+0^2+(-2)^2}}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{9\times 2+k\times 0+(-3)\times(-2)}{\sqrt{90+k^2}\times\sqrt{8}}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{24}{\sqrt{720+8k^2}}$$

$$\Rightarrow \sqrt{2160+24k^2} = 48$$

$$\Rightarrow 2160+24k^2 = 2304$$

$$\Rightarrow k^2 = 6 \Rightarrow k = \pm\sqrt{6}$$

As k is a positive constant, $k = \sqrt{6}$

21)

$$\cos 60° = \frac{\begin{pmatrix} 8 \\ 1 \\ 2k \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -2 \\ 0 \end{pmatrix}}{\sqrt{8^2+1^2+(2k)^2} \times \sqrt{4^2+(-2)^2+0^2}}$$

$$\Rightarrow \frac{1}{2} = \frac{8\times4+1\times(-2)+2k\times0}{\sqrt{4k^2+65}\times\sqrt{20}}$$

$$\Rightarrow \frac{1}{2} = \frac{30}{\sqrt{80k^2+1300}}$$

$$\Rightarrow \sqrt{80k^2+1300} = 60$$

$$\Rightarrow 80k^2+1300 = 3600$$

$$\Rightarrow k^2 = \frac{115}{4} \Rightarrow k = \pm\frac{\sqrt{115}}{2}$$

As k is a negative constant, $k = -\frac{\sqrt{115}}{2}$

22)

$$\frac{1}{9}=\frac{\begin{pmatrix}8\\-4\\1\end{pmatrix}\cdot\begin{pmatrix}4\\k\\-2\end{pmatrix}}{\sqrt{8^2+(-4)^2+1^2}\times\sqrt{4^2+k^2+(-2)^2}}$$

$$\Rightarrow \frac{1}{9}=\frac{8\times4+(-4)\times k+1\times(-2)}{9\times\sqrt{20+k^2}}$$

$$\Rightarrow 1=\frac{30-4k}{\sqrt{20+k^2}}$$

$$\Rightarrow \sqrt{20+k^2}=30-4k$$

$$\Rightarrow 20+k^2=(30-4k)^2$$

$$\Rightarrow 20+k^2=900-240k+16k^2$$

$$\Rightarrow 0=15k^2-240k+880$$

$$\Rightarrow 0=3k^2-48k+176$$

$$\Rightarrow 0=3(k^2-16k)+176$$

$$\Rightarrow 0=3((k-8)^2-64)+176$$

$$\Rightarrow 0=3(k-8)^2-192+176$$

$$\Rightarrow 0=3(k-8)^2-16$$

$$\Rightarrow (k-8)^2=\frac{16}{3}$$

$$\Rightarrow k-8=\pm\frac{4}{\sqrt{3}}$$

$$\Rightarrow k=8\pm\frac{4}{\sqrt{3}}$$

Solution must solve $\sqrt{20+k^2}=30-4k$

If $k=8+\frac{4}{\sqrt{3}}$ then $30-4k=-\frac{6+16\sqrt{3}}{3}<0$

If $k=8-\frac{4}{\sqrt{3}}$ then $30-4k=\frac{-6+16\sqrt{3}}{3}>0$

$$\therefore k=8-\frac{4}{\sqrt{3}}$$

23)

Π_1 has equation $10x - 5y + pz = d_1$

Π_2 has equation $4x + 4y - 8z = d_2$

$(2, 1, -5)$ lies on Π_1 so:

$10(2) - 5(1) + p(-5) = d_1 \Rightarrow 15 - 5p = d_1$

The angle between Π_1 and Π_2 is $60°$ so:

$$\cos 60° = \frac{\begin{pmatrix} 10 \\ -5 \\ p \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 4 \\ -8 \end{pmatrix}}{\sqrt{10^2+(-5)^2+p^2} \times \sqrt{4^2+4^2+(-8)^2}}$$

$$\Rightarrow \frac{1}{2} = \frac{10\times4+(-5)\times4+p\times(-8)}{\sqrt{125+p^2}\times\sqrt{96}}$$

$$\Rightarrow \frac{1}{2} = \frac{20-8p}{\sqrt{12000+96p^2}}$$

$$\Rightarrow \sqrt{12000 + 96p^2} = 40 - 16p$$

$$\Rightarrow 12000 + 96p^2 = (40 - 16p)^2$$

$$\Rightarrow 12000 + 96p^2 = 1600 - 1280p + 256p^2$$

$$\Rightarrow 0 = 160p^2 - 1280p - 10400$$

$$\Rightarrow 0 = p^2 - 8p - 65$$

$$\Rightarrow 0 = (p - 13)(p + 5)$$

$\Rightarrow p = 13$ or $p = -5$

As p is a negative constant, $p = -5$

$\Rightarrow d_1 = 15 - 5(-5) = 40$

$\therefore$ The cartesian equation of Π_1 is $10x - 5y - 5z = 40 \Rightarrow 2x - y - z = 8$

24)

If $\mathbf{m}$ lies in the plane, then

$\begin{pmatrix}3\\0\\1\end{pmatrix}\cdot\begin{pmatrix}p\\q\\r\end{pmatrix}=0\Rightarrow 3p+r=0\Rightarrow r=-3p$

$|\mathbf{m}|=4\Rightarrow\sqrt{p^2+q^2+r^2}=4\Rightarrow p^2+q^2+r^2=16$

$\cos 45°=\frac{\begin{pmatrix}p\\q\\r\end{pmatrix}\cdot\begin{pmatrix}0\\1\\0\end{pmatrix}}{\sqrt{p^2+q^2+r^2}\times\sqrt{0^2+1^2+0^2}}$

$\Rightarrow\frac{\sqrt{2}}{2}=\frac{p\times0+q\times1+r\times0}{4\times1}\Rightarrow\frac{\sqrt{2}}{2}=\frac{q}{4}\Rightarrow q=2\sqrt{2}$

$\therefore p^2+\left(2\sqrt{2}\right)^2+r^2=16\Rightarrow p^2+r^2=8$

Now solve the simultaneous equations $r=-3p$ and $p^2+r^2=8$:

$p^2+(-3p)^2=8\Rightarrow p^2+9p^2=8\Rightarrow 10p^2=8\Rightarrow p^2=\frac{4}{5}\Rightarrow p=\pm\frac{2\sqrt{5}}{5}$

As p is a positive constant, $p=\frac{2\sqrt{5}}{5}$

$\therefore r=-3\left(\frac{2\sqrt{5}}{5}\right)=-\frac{6\sqrt{5}}{5}$

So $\mathbf{m}=\frac{2\sqrt{5}}{5}\mathbf{i}+2\sqrt{2}\mathbf{j}-\frac{6\sqrt{5}}{5}\mathbf{k}$

25)

If $\mathbf{p}$ lies in the plane, then

$\begin{pmatrix}12\\-4\\0\end{pmatrix}\cdot\begin{pmatrix}a\\b\\c\end{pmatrix}=0\Rightarrow 12a-4b=0\Rightarrow b=3a$

$|\mathbf{p}|=6\sqrt{10}\Rightarrow\sqrt{a^2+b^2+c^2}=6\sqrt{10}\Rightarrow a^2+b^2+c^2=360$

$\cos 30°=\frac{\begin{pmatrix}a\\b\\c\end{pmatrix}\cdot\begin{pmatrix}0\\0\\1\end{pmatrix}}{\sqrt{a^2+b^2+c^2}\times\sqrt{0^2+0^2+1^2}}$

$\Rightarrow\frac{\sqrt{3}}{2}=\frac{a\times0+b\times0+c\times1}{6\sqrt{10}\times1}\Rightarrow\frac{\sqrt{3}}{2}=\frac{c}{6\sqrt{10}}\Rightarrow c=3\sqrt{30}$

$\therefore a^2+b^2+\left(3\sqrt{30}\right)^2=360\Rightarrow a^2+b^2=90$

Now solve the simultaneous equations $b=3a$ and $a^2+b^2=90$:

$a^2+(3a)^2=90\Rightarrow a^2+9a^2=90\Rightarrow 10a^2=90\Rightarrow a^2=9\Rightarrow a=\pm3$

As a is a positive constant, $a=3$

$\therefore b=3(3)=9$

So $\mathbf{p}=3\mathbf{i}+9\mathbf{j}+3\sqrt{30}\mathbf{k}$

26)

Let $\mathbf{v} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$

If $\mathbf{v}$ is a unit vector then $\sqrt{a^2 + b^2 + c^2} = 1 \Rightarrow a^2 + b^2 + c^2 = 1$

If $\mathbf{v}$ is parallel to the plane $x - 2y = 5$ then:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} = 0 \Rightarrow a - 2b = 0 \Rightarrow b = \frac{1}{2}a$$

$$\cos 45° = \frac{\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} 8 \\ -3 \\ 5 \end{pmatrix}}{\sqrt{a^2+b^2+c^2} \times \sqrt{8^2+(-3)^2+5^2}}$$

$$\Rightarrow \frac{\sqrt{2}}{2} = \frac{a \times 8 + b \times (-3) + c \times 5}{1 \times \sqrt{98}} \Rightarrow \frac{\sqrt{2}}{2} = \frac{8a - 3b + 5c}{7\sqrt{2}} \Rightarrow 8a - 3b + 5c = 7$$

$$8a - 3\left(\frac{1}{2}a\right) + 5c = 7 \Rightarrow 5c = 7 - \frac{13}{2}a \Rightarrow c = \frac{7}{5} - \frac{13}{10}a$$

$$a^2 + \left(\frac{1}{2}a\right)^2 + \left(\frac{7}{5} - \frac{13}{10}a\right)^2 = 1$$

$$\Rightarrow a^2 + \frac{1}{4}a^2 + \frac{169}{100}a^2 - \frac{91}{25}a + \frac{49}{25} = 1 \Rightarrow \frac{147}{50}a^2 - \frac{91}{25}a + \frac{24}{25} = 0$$

$$\Rightarrow 147a^2 - 182a + 48 = 0 \Rightarrow (21a - 8)(7a - 6) = 0$$

$\Rightarrow a = \frac{8}{21}$ or $a = \frac{6}{7}$

If $a = \frac{8}{21}$ then $b = \frac{4}{21}$ and $c = \frac{19}{21}$, so $\mathbf{v} = \frac{8}{21}\mathbf{i} + \frac{4}{21}\mathbf{j} + \frac{19}{21}\mathbf{k}$

If $a = \frac{6}{7}$ then $b = \frac{3}{7}$ and $c = \frac{2}{7}$, so $\mathbf{v} = \frac{6}{7}\mathbf{i} + \frac{3}{7}\mathbf{j} + \frac{2}{7}\mathbf{k}$

27) a)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 9 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ -5 \end{pmatrix} = \begin{pmatrix} 7 \\ -2 \\ 5 \end{pmatrix}$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} -7 \\ 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ -5 \end{pmatrix} = \begin{pmatrix} -9 \\ 0 \\ 7 \end{pmatrix}$$

27) b)

$$\begin{pmatrix} 7 \\ -2 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 7 \\ 47 \\ 9 \end{pmatrix} = 7 \times 7 + (-2) \times 47 + 5 \times 9 = 49 - 94 + 45 = 0$$

$$\begin{pmatrix} -9 \\ 0 \\ 7 \end{pmatrix} \cdot \begin{pmatrix} 7 \\ 47 \\ 9 \end{pmatrix} = (-9) \times 7 + 0 \times 47 + 7 \times 9 = -63 + 0 + 63 = 0$$

$\therefore$ both $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are perpendicular to $7\mathbf{i} + 47\mathbf{j} + 9\mathbf{k}$

27) c)

The plane has equation $7x + 47y + 9z = d$

$d = 7(2) + 47(3) + 9(-5) = 110$

$\therefore 7x + 47y + 9z = 110$

28) a)

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \begin{pmatrix} 3 \\ 2 \\ 8 \end{pmatrix} - \begin{pmatrix} -1 \\ 2 \\ 6 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix}$$

$$\overrightarrow{PR} = \overrightarrow{OR} - \overrightarrow{OP} = \begin{pmatrix} 4 \\ -3 \\ -5 \end{pmatrix} - \begin{pmatrix} -1 \\ 2 \\ 6 \end{pmatrix} = \begin{pmatrix} 5 \\ -5 \\ -11 \end{pmatrix}$$

28) b)

$$\begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 27 \\ -10 \end{pmatrix} = 4 \times 5 + 0 \times 27 + 2 \times (-10) = 20 + 0 - 20 = 0$$

$$\begin{pmatrix} 5 \\ -5 \\ -11 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 27 \\ -10 \end{pmatrix} = 5 \times 5 + (-5) \times 27 + (-11) \times (-10) = 25 - 135 + 110 = 0$$

$\therefore$ both $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ are perpendicular to $5\mathbf{i} + 27\mathbf{j} - 10\mathbf{k}$

28) c)

The plane has equation $5x + 27y - 10z = d$

$d = 5(-1) + 27(2) - 10(6) = -11$

$\therefore 5x + 27y - 10z = -11$

29) a)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 6 \\ -1 \\ 0 \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \\ -2 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix}$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \times \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 2 \\ -2 & 1 & -1 \end{vmatrix}$$

$$= \big((-1) \times (-1) - 2 \times 1\big)\mathbf{i} - \big(2 \times (-1) - 2 \times (-2)\big)\mathbf{j} + \big(2 \times 1 - (-1) \times (-2)\big)\mathbf{k}$$

$$= -\mathbf{i} - 2\mathbf{j}$$

29) b)

The plane has equation $-x-2y=d$

$d=-(4)-2(0)+0(-2)=-4$

$\therefore -x-2y=-4$

$\Rightarrow x+2y=4$

30)

Let A, B and C be $(2,1,1)$, $(0,1,3)$ and $(1,0,4)$ respectively.

$$\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}=\begin{pmatrix}0\\1\\3\end{pmatrix}-\begin{pmatrix}2\\1\\1\end{pmatrix}=\begin{pmatrix}-2\\0\\2\end{pmatrix}$$

$$\overrightarrow{AC}=\overrightarrow{OC}-\overrightarrow{OA}=\begin{pmatrix}1\\0\\4\end{pmatrix}-\begin{pmatrix}2\\1\\1\end{pmatrix}=\begin{pmatrix}-1\\-1\\3\end{pmatrix}$$

$$\overrightarrow{AB}\times\overrightarrow{AC}=\begin{pmatrix}-2\\0\\2\end{pmatrix}\times\begin{pmatrix}-1\\-1\\3\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-2&0&2\\-1&-1&3\end{vmatrix}$$

$$=\big(0\times3-2\times(-1)\big)\mathbf{i}-\big((-2)\times3-2\times(-1)\big)\mathbf{j}+\big((-2)\times(-1)-0\times(-1)\big)\mathbf{k}$$

$$=2\mathbf{i}+4\mathbf{j}+2\mathbf{k}$$

The plane has equation $2x+4y+2z=d$

$d=2(2)+4(1)+2(1)=10$

$\therefore 2x+4y+2z=10$

$\Rightarrow x+2y+z=5$

31) a)

Let A, B and C be $(1,0,1)$, $(1,1,1)$ and $(0,1,0)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}1\\1\\1\end{pmatrix} - \begin{pmatrix}1\\0\\1\end{pmatrix} = \begin{pmatrix}0\\1\\0\end{pmatrix}$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix}0\\1\\0\end{pmatrix} - \begin{pmatrix}1\\0\\1\end{pmatrix} = \begin{pmatrix}-1\\1\\-1\end{pmatrix}$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix}0\\1\\0\end{pmatrix} \times \begin{pmatrix}-1\\1\\-1\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\0 & 1 & 0\\-1 & 1 & -1\end{vmatrix}$$

$$= (1 \times (-1) - 0 \times 1)\mathbf{i} - \big(0 \times (-1) - 0 \times (-1)\big)\mathbf{j} + \big(0 \times 1 - 1 \times (-1)\big)\mathbf{k}$$

$$= -\mathbf{i} + \mathbf{k}$$

The plane has equation $-x + z = d$

$d = -1(1) + 0(0) + 1(1) = 0$

$\therefore -x + z = 0$

31) b)

Let A, B and C be $(2,1,1)$, $(-1,1,2)$ and $(3,1,-1)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}-1\\1\\2\end{pmatrix} - \begin{pmatrix}2\\1\\1\end{pmatrix} = \begin{pmatrix}-3\\0\\1\end{pmatrix}$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix}3\\-1\\-1\end{pmatrix} - \begin{pmatrix}2\\1\\1\end{pmatrix} = \begin{pmatrix}1\\-2\\-2\end{pmatrix}$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix}-3\\0\\1\end{pmatrix} \times \begin{pmatrix}1\\-2\\-2\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\-3 & 0 & 1\\1 & -2 & -2\end{vmatrix}$$

$$= \big(0 \times (-2) - 1 \times (-2)\big)\mathbf{i} - \big((-3) \times (-2) - 1 \times 1\big)\mathbf{j} + \big((-3) \times (-2) - 0 \times 1\big)\mathbf{k}$$

$$= 2\mathbf{i} - 5\mathbf{j} + 6\mathbf{k}$$

The plane has equation $2x - 5y + 6z = d$

$d = 2(2) - 5(1) + 6(1) = 5$

$\therefore 2x - 5y + 6z = 5$

31) c)

Let A, B and C be $(-1, 4, 5)$, $(3, 2, -5)$ and $(4, 2, -3)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \\ -10 \end{pmatrix}$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 4 \\ 2 \\ -3 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 5 \\ -2 \\ -8 \end{pmatrix}$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} 4 \\ -2 \\ -10 \end{pmatrix} \times \begin{pmatrix} 5 \\ -2 \\ -8 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -2 & -10 \\ 5 & -2 & -8 \end{vmatrix}$$

$$= \big((-2) \times (-8) - (-10) \times (-2)\big)\mathbf{i} - (4 \times (-8) - (-10) \times 5)\mathbf{j} + (4 \times (-2) - (-2) \times 5)\mathbf{k}$$

$$= -4\mathbf{i} - 18\mathbf{j} + 2\mathbf{k}$$

The plane has equation $-4x - 18y + 2z = d$

$d = -4(-1) - 18(4) + 2(5) = -58$

$\therefore -4x - 18y + 2z = -58$

$\Rightarrow 2x + 9y - z = 29$

31) d)

Let A, B and C be $(10, -20, 30)$, $(-20, 40, 50)$ and $(0, 0, -10)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} -20 \\ 40 \\ 50 \end{pmatrix} - \begin{pmatrix} 10 \\ -20 \\ 30 \end{pmatrix} = \begin{pmatrix} -30 \\ 60 \\ 20 \end{pmatrix}$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 0 \\ 0 \\ -10 \end{pmatrix} - \begin{pmatrix} 10 \\ -20 \\ 30 \end{pmatrix} = \begin{pmatrix} -10 \\ 20 \\ -40 \end{pmatrix}$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} -30 \\ 60 \\ 20 \end{pmatrix} \times \begin{pmatrix} -10 \\ 20 \\ -40 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -30 & 60 & 20 \\ -10 & 20 & -40 \end{vmatrix}$$

$$= (60 \times (-40) - 20 \times 20)\mathbf{i} - \big((-30) \times (-40) - 20 \times (-10)\big)\mathbf{j} + \big((-30) \times 20 - 60 \times (-10)\big)\mathbf{k}$$

$$= -2800\mathbf{i} - 1400\mathbf{j}$$

The plane has equation $-2800x - 1400y = d$

$d = -2800(10) - 1400(-20) = 0$

$\therefore -2800x - 1400y = 0$

$\Rightarrow 2x + y = 0$

32) a)

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 4 \\ 1 \\ -2 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$

$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 2 \\ 0 \\ 6 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ 11 \end{pmatrix}$

$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} \times \begin{pmatrix} -1 \\ -2 \\ 11 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 3 \\ -1 & -2 & 11 \end{vmatrix}$

$= \big((-1) \times 11 - 3 \times (-2)\big)\mathbf{i} - \big(1 \times 11 - 3 \times (-1)\big)\mathbf{j} + \big(1 \times (-2) - (-1) \times (-1)\big)\mathbf{k}$

$= -5\mathbf{i} - 14\mathbf{j} - 3\mathbf{k}$

The plane has equation $-5x - 14y - 3z = d$

$d = -5(3) - 14(2) - 3(-5) = -28$

$\therefore -5x - 14y - 3z = -28$

$\Rightarrow 5x + 14y + 3z = 28$

$5(k) + 14(1) + 3(2) = 28 \Rightarrow 5k + 20 = 28 \Rightarrow k = \frac{8}{5}$

32) b)

$5(3) + 14(k) + 3(2) = 28 \Rightarrow 21 + 14k = 28 \Rightarrow k = \frac{1}{2}$

32) c)

$5(0) + 14(5) + 3(k) = 28 \Rightarrow 70 + 3k = 28 \Rightarrow k = -14$

33)

$\begin{pmatrix} 2 & -3 & k \\ 4 & 1 & -10 \\ -2 & -11 & 5 \end{pmatrix} \begin{pmatrix} p \\ q \\ r \end{pmatrix} = \begin{pmatrix} 2p - 3q + kr \\ 4p + q - 10r \\ -2p - 11q + 5r \end{pmatrix}$

Substitute these into the equation of the plane:

$3(2p - 3q + kr) - 2(4p + q - 10r) - (-2p - 11q + 5r) = 0$

$\Rightarrow 6p - 9q + 3kr - 8p - 2q + 20r + 2p + 11q - 5r = 0$

$\Rightarrow 3kr + 15r = 0$

$\Rightarrow k = -5$

34) a)

$$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = \begin{pmatrix}1\\3\\4\end{pmatrix} - \begin{pmatrix}0\\1\\3\end{pmatrix} = \begin{pmatrix}1\\2\\1\end{pmatrix}$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}2\\0\\1\end{pmatrix} - \begin{pmatrix}0\\1\\3\end{pmatrix} = \begin{pmatrix}2\\-1\\-2\end{pmatrix}$$

$$\overrightarrow{AD} \times \overrightarrow{AB} = \begin{pmatrix}1\\2\\1\end{pmatrix} \times \begin{pmatrix}2\\-1\\-2\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\1 & 2 & 1\\2 & -1 & -2\end{vmatrix}$$

$$= \big(2 \times (-2) - 1 \times (-1)\big)\mathbf{i} - (1 \times (-2) - 1 \times 2)\mathbf{j} + (1 \times (-1) - 2 \times 2)\mathbf{k}$$

$$= -3\mathbf{i} + 4\mathbf{j} - 5\mathbf{k}$$

The plane has equation $-3x + 4y - 5z = d$

$d = -3(0) + 4(1) - 5(3) = -11$

$\therefore -3x + 4y - 5z = -11$

$\Rightarrow 3x - 4y + 5z = 11$

34) b)

$3(4) - 4(1) + 5(k) = 11 \Rightarrow 8 + 5k = 11 \Rightarrow k = \frac{3}{5}$

34) c)

$$\cos\theta = \frac{\begin{pmatrix}3\\-4\\5\end{pmatrix} \cdot \begin{pmatrix}0\\0\\1\end{pmatrix}}{\sqrt{3^2+(-4)^2+5^2} \times \sqrt{0^2+0^2+1^2}}$$

$$\Rightarrow \cos\theta = \frac{3\times0+(-4)\times0+5\times1}{\sqrt{50}\times\sqrt{1}}$$

$$\Rightarrow \cos\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = 45°$$

Acute Angle $= 45°$

35)

First find Π_1:

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 2 \\ 0 \\ -2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$

$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix}$

$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} \times \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -2 & -1 \\ -1 & -1 & 2 \end{vmatrix}$

$= \big((-2) \times 2 - (-1) \times (-1)\big)\mathbf{i} - \big(1 \times 2 - (-1) \times (-1)\big)\mathbf{j} + \big(1 \times (-1) - (-2) \times (-1)\big)\mathbf{k}$

$= -5\mathbf{i} - \mathbf{j} - 3\mathbf{k}$

The plane has equation $-5x - y - 3z = d$

$d = -5(1) - (2) - 3(-1) = -4$

$\therefore -5x - y - 3z = -4$

$\Rightarrow \Pi_1: 5x + y + 3z = 4$

Now find Π_2:

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 2 \\ 0 \\ -2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$

$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix}$

$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} \times \begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -2 & -1 \\ 2 & -4 & 2 \end{vmatrix}$

$= \big((-2) \times 2 - (-1) \times (-4)\big)\mathbf{i} - (1 \times 2 - (-1) \times 2)\mathbf{j} + (1 \times (-4) - (-2) \times 2)\mathbf{k}$

$= -8\mathbf{i} - 4\mathbf{j}$

The plane has equation $-8x - 4y = d$

$d = -8(1) - 4(2) = -16$

$\therefore -8x - 4y = -16$

$\Rightarrow \Pi_2: 2x + y = 4$

Now find the acute angle between Π_1 and Π_2:

$\cos\theta = \frac{\begin{pmatrix} 5 \\ 1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}}{\sqrt{5^2+1^2+3^2} \times \sqrt{2^2+1^2+0^2}}$

$\Rightarrow \cos\theta = \frac{5\times2+1\times1+3\times0}{\sqrt{35}\sqrt{5}}$

$\Rightarrow \cos\theta = \frac{11}{5\sqrt{7}} = \frac{11\sqrt{7}}{35} \Rightarrow \theta = 33.744\ldots°$

Acute Angle = 33.7° (1 DP)

36) a)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} \frac{\sqrt{15}}{2} \\ 0 \\ -\frac{\sqrt{5}}{2} \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ \sqrt{5} \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{15}}{2} \\ 0 \\ -\frac{3\sqrt{5}}{2} \end{pmatrix}$$

$$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = \begin{pmatrix} 0 \\ \sqrt{10} \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ \sqrt{5} \end{pmatrix} = \begin{pmatrix} 0 \\ \sqrt{10} \\ -\sqrt{5} \end{pmatrix}$$

$$\overrightarrow{AB} \times \overrightarrow{AD} = \begin{pmatrix} \frac{\sqrt{15}}{2} \\ 0 \\ -\frac{3\sqrt{5}}{2} \end{pmatrix} \times \begin{pmatrix} 0 \\ \sqrt{10} \\ -\sqrt{5} \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\sqrt{15}}{2} & 0 & -\frac{3\sqrt{5}}{2} \\ 0 & \sqrt{10} & -\sqrt{5} \end{vmatrix}$$

$$= \left(0 \times \left(-\sqrt{5}\right) - \left(-\frac{3\sqrt{5}}{2}\right) \times \sqrt{10}\right)\mathbf{i} - \left(\frac{\sqrt{15}}{2} \times \left(-\sqrt{5}\right) - \left(-\frac{3\sqrt{5}}{2}\right) \times 0\right)\mathbf{j} + \left(\frac{\sqrt{15}}{2} \times \sqrt{10} - 0 \times 0\right)\mathbf{k}$$

$$= \frac{15\sqrt{2}}{2}\mathbf{i} + \frac{5\sqrt{3}}{2}\mathbf{j} + \frac{5\sqrt{6}}{2}\mathbf{k}$$

The plane has equation $\frac{15\sqrt{2}}{2}x + \frac{5\sqrt{3}}{2}y + \frac{5\sqrt{6}}{2}z = d$

$$d = \frac{15\sqrt{2}}{2}(0) + \frac{5\sqrt{3}}{2}(0) + \frac{5\sqrt{6}}{2}\left(\sqrt{5}\right) = \frac{5\sqrt{30}}{2}$$

$$\therefore \frac{15\sqrt{2}}{2}x + \frac{5\sqrt{3}}{2}y + \frac{5\sqrt{6}}{2}z = \frac{5\sqrt{30}}{2}$$

$$\Rightarrow 3\sqrt{2}x + \sqrt{3}y + \sqrt{6}z = \sqrt{30}$$

36) b)

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} -\frac{\sqrt{15}}{2} \\ 0 \\ -\frac{\sqrt{5}}{2} \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ \sqrt{5} \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{15}}{2} \\ 0 \\ -\frac{3\sqrt{5}}{2} \end{pmatrix}$$

$$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = \begin{pmatrix} 0 \\ \sqrt{10} \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ \sqrt{5} \end{pmatrix} = \begin{pmatrix} 0 \\ \sqrt{10} \\ -\sqrt{5} \end{pmatrix}$$

$$\overrightarrow{AC} \times \overrightarrow{AD} = \begin{pmatrix} -\frac{\sqrt{15}}{2} \\ 0 \\ -\frac{3\sqrt{5}}{2} \end{pmatrix} \times \begin{pmatrix} 0 \\ \sqrt{10} \\ -\sqrt{5} \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\frac{\sqrt{15}}{2} & 0 & -\frac{3\sqrt{5}}{2} \\ 0 & \sqrt{10} & -\sqrt{5} \end{vmatrix}$$

$$= \left(0 \times \left(-\sqrt{5}\right) - \left(-\frac{3\sqrt{5}}{2}\right) \times \sqrt{10}\right)\mathbf{i} - \left(\left(-\frac{\sqrt{15}}{2}\right) \times \left(-\sqrt{5}\right) - \left(-\frac{3\sqrt{5}}{2}\right) \times 0\right)\mathbf{j} +$$

$$\left(\left(-\frac{\sqrt{15}}{2}\right) \times \sqrt{10} - 0 \times 0\right)\mathbf{k}$$

$$= \frac{15\sqrt{2}}{2}\mathbf{i} - \frac{5\sqrt{3}}{2}\mathbf{j} - \frac{5\sqrt{6}}{2}\mathbf{k}$$

The plane has equation $\frac{15\sqrt{2}}{2}x - \frac{5\sqrt{3}}{2}y - \frac{5\sqrt{6}}{2}z = d$

$$d = \frac{15\sqrt{2}}{2}(0) - \frac{5\sqrt{3}}{2}(0) - \frac{5\sqrt{6}}{2}\left(\sqrt{5}\right) = -\frac{5\sqrt{30}}{2}$$

$$\therefore \frac{15\sqrt{2}}{2}x - \frac{5\sqrt{3}}{2}y - \frac{5\sqrt{6}}{2}z = -\frac{5\sqrt{30}}{2}$$

$$\Rightarrow 3\sqrt{2}x - \sqrt{3}y - \sqrt{6}z = -\sqrt{30}$$

36) c)

$$\cos\theta = \frac{\begin{pmatrix} 3\sqrt{2} \\ \sqrt{3} \\ \sqrt{6} \end{pmatrix} \cdot \begin{pmatrix} 3\sqrt{2} \\ -\sqrt{3} \\ -\sqrt{6} \end{pmatrix}}{\sqrt{(3\sqrt{2})^2 + (\sqrt{3})^2 + (\sqrt{6})^2} \times \sqrt{(3\sqrt{2})^2 + (-\sqrt{3})^2 + (-\sqrt{6})^2}}$$

$$\Rightarrow \cos\theta = \frac{3\sqrt{2} \times 3\sqrt{2} + \sqrt{3} \times (-\sqrt{3}) + \sqrt{6} \times (-\sqrt{6})}{\sqrt{27} \times \sqrt{27}}$$

$$\Rightarrow \cos\theta = \frac{9}{27} = \frac{1}{3} \Rightarrow \theta = 70.528\ldots°$$

Acute angle is 70.5° (1 DP)